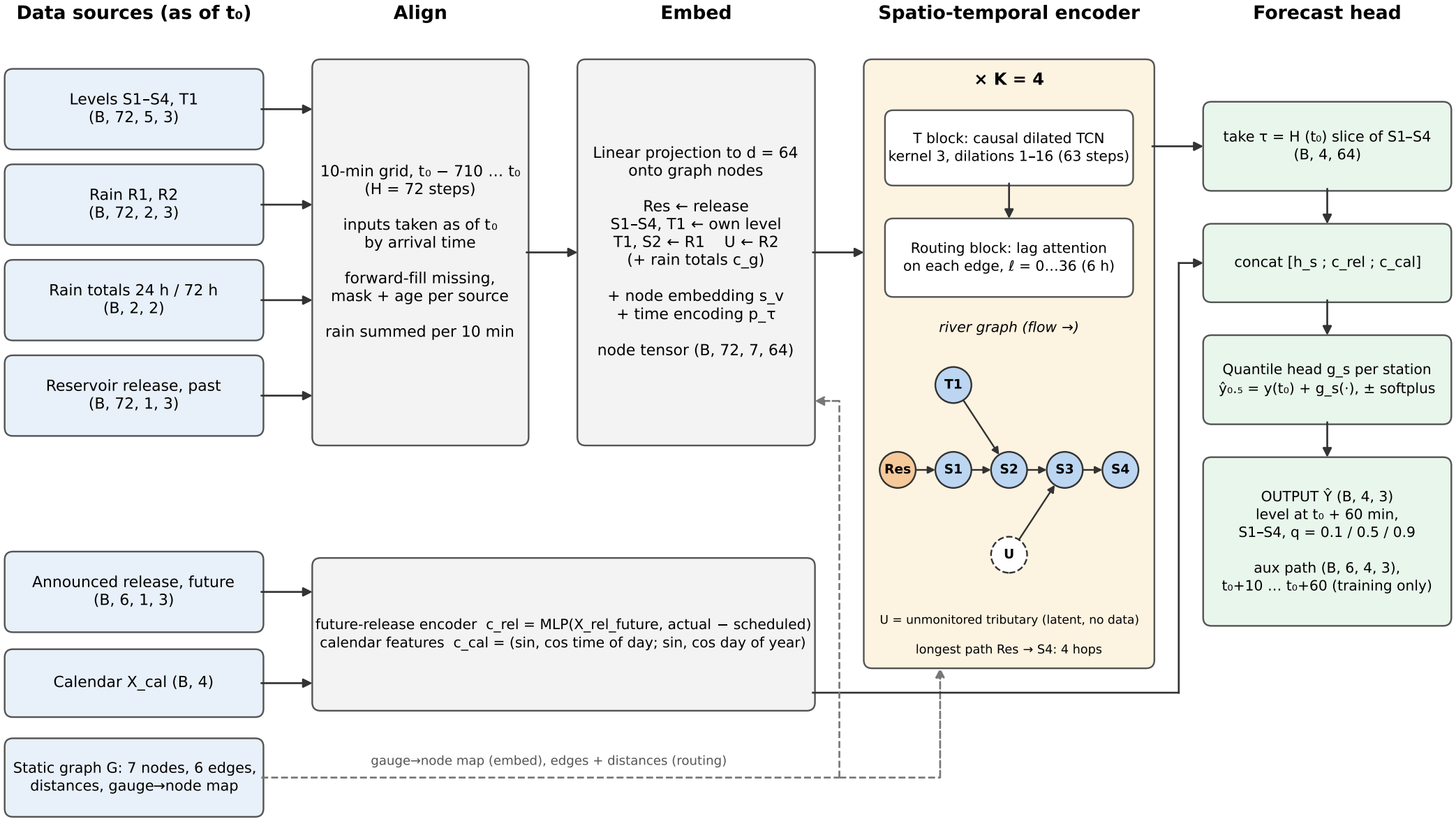
Data sources (as of t₀)
Align
Embed
Spatio-temporal encoder
Forecast head
Levels S1–S4, T1
(B, 72, 5, 3)
Rain R1, R2
(B, 72, 2, 3)
Rain totals 24 h / 72 h
(B, 2, 2)
Reservoir release, past
(B, 72, 1, 3)
10-min grid, t₀ − 710 … t₀
(H = 72 steps)
inputs taken as of t₀
by arrival time
forward-fill missing,
mask + age per source
rain summed per 10 min
Linear projection to d = 64
onto graph nodes
Res ← release
S1–S4, T1 ← own level
T1, S2 ← R1 U ← R2
(+ rain totals c_g)
+ node embedding s_v
+ time encoding p_τ
node tensor (B, 72, 7, 64)
× K = 4
T block: causal dilated TCN
kernel 3, dilations 1–16 (63 steps)
Routing block: lag attention
on each edge, ℓ = 0…36 (6 h)
river graph (flow →)
T1
Res
S1
S2
S3
S4
U
U = unmonitored tributary (latent, no data)
longest path Res → S4: 4 hops
take τ = H (t₀) slice of S1–S4
(B, 4, 64)
concat [h_s ; c_rel ; c_cal]
Quantile head g_s per station
ŷ₀.₅ = y(t₀) + g_s(·), ± softplus
OUTPUT Ŷ (B, 4, 3)
level at t₀ + 60 min,
S1–S4, q = 0.1 / 0.5 / 0.9
aux path (B, 6, 4, 3),
t₀+10 … t₀+60 (training only)
Announced release, future
(B, 6, 1, 3)
Calendar X_cal (B, 4)
future-release encoder c_rel = MLP(X_rel_future, actual − scheduled)
calendar features c_cal = (sin, cos time of day; sin, cos day of year)
Static graph G: 7 nodes, 6 edges,
distances, gauge→node map
gauge→node map (embed), edges + distances (routing)